

Probability and Statistics

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1 Chapter 1 - Samples, Spaces and Events

1.1 Definitions

Definition 1. *The sample space is the set of all possible outcomes for the experiment. It is denoted by S (or Ω).*

Definition 2. *An event is a subset of the sample space. The event occurs if the actual outcome is an element of this subset.*

Definition 3. *An event E is a simple event (or elementary event) if it consists of a single element of the sample space S .*

1.2 Notes

The size, or cardinality, of a finite set is the number of elements it contains.

- The set difference $A \setminus B$ (“ A take away B ”) is the set of elements in A but not in B :

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

- The symmetric difference $A \triangle B$ is the set of elements in either A or B , but not both:

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

- The complement of a set A is the set of elements in the sample space that are not in A . It can be denoted by A' or A^c :

$$A^c = S \setminus A.$$

- Two sets A and B are disjoint (or mutually exclusive) if they have no elements in common:

$$A \cap B = \emptyset.$$

De Morgan’s laws describe how complements interact with unions and intersections:

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

In words, the complement of a union is the intersection of the complements, and the complement of an intersection is the union of the complements.

2 Chapter 2

2.1 Definitions

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19.1 **Definitions**

19.2 **Notes**

20 Chapter 20

20.1 **Definitions**

20.2 **Notes**